VINEETH KALLURU

Applied AI Engineer | Production LLM Systems, Agents and Post-Training | IEEE Senior Member

San Francisco Bay Area | vineethkalluru@gmail.com | vineethkalluru.com | linkedin.com/in/vineethkalluru | github.com/dreamhunter02 | Google Scholar

SUMMARY

Applied AI engineer who takes open models and agents from POC to production inside enterprise environments. At NVIDIA, the technical lead for 25+ Tier-1 enterprise software and industrial partners plus hyperscale cloud and frontier-lab partners: solution architecture, evals, post-training data, inference sizing and tuning, agent harnesses, guardrails, and executive advisory. Diagnoses production-blocking failures, ships the fix, and turns it into product changes. Earlier work on TPU performance modeling at Google and a dataflow AI compiler at SambaNova. Co-author of the NVIDIA Nemotron technical reports; first author on agentic-AI papers at IEEE ICA 2026 and IEEE ISSRE 2026.

EXPERIENCE

Senior Solutions Architect, Generative AI, NVIDIA (Strategic ISV AI Foundry, Santa Clara, CA) **Nov 2023 - 2026**

- **Solution design, POC to production:** Owned end-to-end architecture for enterprise search, collaboration, data-management and document-AI platforms: RAG, agentic workflows, guardrails and model routing on cloud, on-prem and air-gapped GPU deployments. Took open models from evaluation to GA production, including a small model behind a partner's production LLM router and two partner launches featured at GTC 2026.
- **Inference at scale:** Architected multi-node and disaggregated LLM serving on H100, H200, GB200 NVL72 and GB300 NVL72 with TensorRT-LLM, NVIDIA Dynamo, NIM, vLLM and Run:ai; built GPU sizing tools reused across the field. NVIDIA technical lead for a hyperscale cloud partner's MLPerf Inference v4.0 submission (46% gain on LLaMA-2 70B).
- **Post-training and data:** Designed post-training data and fine-tuning plans with partner ML teams (domain-specific SLMs, multilingual embedding fine-tunes, SDG before SFT); ran a citation-failure root-cause study whose findings fed Nemotron post-training; led Nemotron fine-tuning workshops.
- **Evals and agent harnesses:** Built reusable eval harnesses for RAG retrieval, document parsing and agent tool-calling that partners adopted as their own methodology; built harnesses that let open models match frontier-model quality on enterprise tasks; built NVIDIA's multi-agent Asset Lifecycle Management reference (NeMo Agent Toolkit).
- **Production debugging:** Root-caused customer-blocking failures (generation loops in a document-parsing VLM, non-determinism under concurrency, GPU-operator and driver blockers, retrieval latency regressions) and shipped fixes partners adopted, such as a decoding guard that removed loop failures before GA.
- **Strategic advisory:** Advised partner product and engineering leadership on model strategy, build-vs-buy and roadmap (for example, reframing a domain-model effort from SFT to context assembly); negotiated NVIDIA's first Evaluation Data License Agreement with a strategic ISV.
- **Executive briefings:** Led the technical track of executive briefings and EBCs with partner C-suites (CEO, CPO, CTO), producing jointly owned 30/60-day technical roadmaps; supported CEO-level escalations and GTC launch announcements.
- **Feedback to product:** Filed about 30 field bugs and RFEs and maintained the consolidated enterprise feedback on Nemotron, NeMo Retriever, NeMo Guardrails and NIM; field requirements I wrote led to a merged NVIDIA Dynamo PR, and several asks shipped as fixes, certified releases or process changes.
- **Open source:** Authored the Nemotron-CC pretraining data pipeline in NeMo Curator (6.7K-line PR), adopted by external foundation-model teams.
- **Multiplier:** Mentored four SAs and an intern, handed off a flagship account with a documented transition, served on SA hiring panels, and co-taught a GTC 2026 DLI workshop (NVIDIA Spot Award).

ML Accelerator Architect, Cloud TPU, Google **Jul 2022 - Nov 2023**

- Performance analysis of Gemini workloads on TPU v6 and v7 compute and memory subsystems; hardware-software co-design under Norm Jouppi and David Patterson.

Compiler / ML Systems Engineer (Team Lead), SambaNova Systems **Feb 2020 - Jul 2022**

- Led a 7-engineer team that shipped the first known production for-loop and RNN support in a dataflow compiler: about 4x faster online inference and 5x faster LSTM training vs. NVIDIA A100 (SambaNova public blog, Aug 2022); about 3x chip-utilization gain from BERT tiling and compiler work.

SELECTED PROJECTS

- **rl-for-ai-research:** GRPO reinforcement-learning training on FinanceBench; open source.
- **WG-Plan:** Workflow-grounded small-model planners for domain-specific tool calling, using workflow graphs to constrain valid actions (research in progress).
- **jarvis-desk-bot:** Hermes agent voice front end on an ESP32 desk robot (local speech, spoken approvals); MIT licensed.

SELECTED PUBLICATIONS

- **Between Weights and Workflows: Executable Graph Contracts for Bounded Enterprise Agents.** IEEE ICA 2026. Accepted (oral); first author.

- **The Coherence Trap: Layer-Decomposed Failure-Mode Discovery in a Deployed Agentic LLM System.** IEEE ISSRE 2026 Industry Track (CORE A). Presented; first author.
- **Beyond the Blue Link** (generative engine optimization in stochastic retrieval) and **Edit-Distance Links** (conditional-tokenization world models). ACM Hypertext 2026; published.
- **NVIDIA Nemotron technical reports:** Nemotron Nano 2 (arXiv:2508.14444, co-author); Llama-Embed-Nemotron-8B (arXiv:2511.07025, contributor).
- **FinanceHarness: Engineering a Finance Retrieval Harness for Search Agents Powered by Sub-10B Models.** RL4LLM-Agents workshop, ACM ICAIF 2026. Accepted; sole author.
- **SkillOpt: Trajectory-Derived, Verifier-Grounded Compilation of LLM-Agent Skills.** ACM CAIS 2026 Agent Skills Workshop; 63% fewer tokens and 62% lower latency at non-regressing reward.
- **HANNA: Hardware Aware Neural Network Architectures using FbNet.** arXiv:1906.07214; first author.

SPEAKING

- NVIDIA GTC 2026 full-day DLI workshop (co-instructor) and poster; GTC DC 2025; Microsoft Build 2024 and Ignite 2024; SC25 demo; SEMI Agentic AI for Semiconductor Manufacturing 2025.
- O'Reilly AI Superstream 2026 ("Building Agent Harnesses to Match Frontier-Model Performance in RAG"); AIKP 2026 invited workshop; TechEquity AI Summit 2026 (Nov 5); CMU guest lecturer (18-740).

RECOGNITION

- **IEEE Senior Member**; IEEE-HKN; NVIDIA NIM Coin nomination and GTC 2026 Spot Award; two granted Indian patents in applied AI/ML (IN 431037, IN 451829); winner, Apple-sponsored Memory Fairness Protocol competition (CMU).

PEER REVIEW (CONFERENCES AND WORKSHOPS)

- **Program committee:** IEEE ICA 2026; KDD 2026 AgenticSE and Agent4IR workshops.
- **Reviewer:** NeurIPS 2026 (ethics review; UserSim and Continual World Models workshops); ACM CAIS 2026 Agent Skills workshop; KDD 2026 workshops (Enterprise AI Agents, AI Data Scientist, SciSoc Agents and LLMs, RelKD); IEEE DSAA 2026; AIKP 2026; IEC2T 2026.

SKILLS

Inference: TensorRT-LLM, NVIDIA Dynamo, NIM, vLLM, Run:ai, disaggregated and multi-node serving, benchmarking (MLPerf)
Models and agents: Nemotron and open LLMs, post-training data design, GRPO, RAG, agent harnesses, NeMo Curator, NeMo Guardrails, NeMo Agent Toolkit, LLM evaluation
Systems and cloud: Python, C/C++, CUDA, ML compilers, Kubernetes; Azure (AzureML, AI Foundry, AKS), AWS, GCP; on-prem and air-gapped

EDUCATION

M.S., Electrical and Computer Engineering, Carnegie Mellon University — **Aug 2018 - Dec 2019**

Professional Program in Artificial Intelligence, Stanford University — **Sep 2022 - Dec 2022**

B.Tech., Electronics and Communication Engineering, SRM Institute of Science and Technology — **Jul 2014 - Apr 2018**